The Brent-Price Causal Effect of the 2026 Hormuz Disruption

A Synthetic Control Approach

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From the previous presentation

[transition recap — to be added]

Problem statement & contribution

The question (part 1).

- Brent price impact of the 2026-02-01 Strait of Hormuz disruption.
- Estimand: ATT over a defined post-event window; method: synthetic control on the observed price series.

What is new in this research field.

- No peer-reviewed SCM applied to a crude benchmark for a geopolitical disruption.
- Closest mechanical precedent: SCM on Austrian retail fuel (Becker, Pfeifer & Schweikert 2021, *Energy Economics*).
- Field default: structural VAR (Kilian 2009; Baumeister & Hamilton 2019) or scenario models (Dallas Fed, OIES, Kiel, IEA).

Contribution and value.

- Data-driven counterfactual complements scenario estimates.
- Method extension: retail fuel $\rightarrow$ crude benchmark; cross-country $\rightarrow$ cross-asset donors; regulatory $\rightarrow$ exogenous geopolitical treatment.
- Policy inputs: SPR sizing, central-bank reaction, fiscal buffers, sovereign hedging, infrastructure investment.

Part 2 of the thesis. [discussed at the end]

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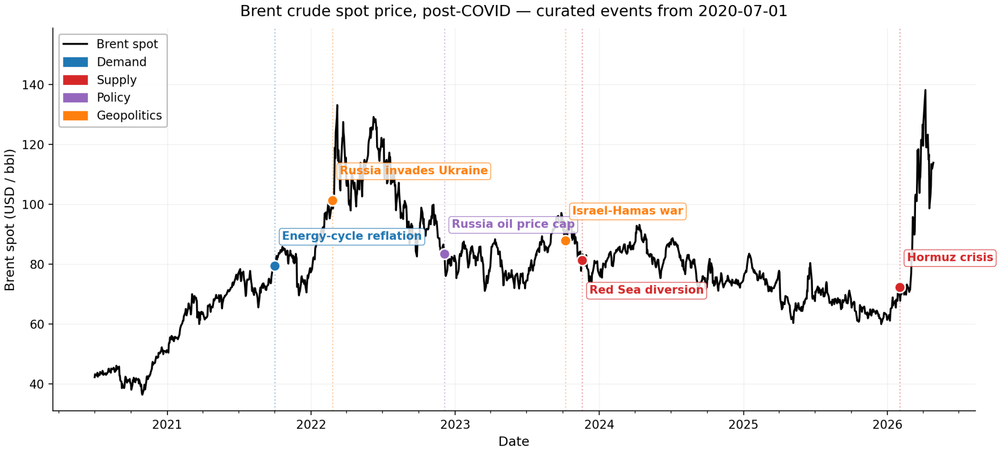


Brent spot, post-COVID. Events: demand (blue), supply (red), policy (purple), geopolitics (orange).

Methodology — ensemble of 5 models

Why an ensemble. Headline = median across models; IQR = model-uncertainty band.

Model	**What it does**	**Reference**
Convex SCM	Convex weights, no extrapolation	Abadie, Diamond & Hainmueller (2010, *JASA*)
Augmented SCM	Convex + ridge bias correction	Ben-Michael, Feller & Rothstein (2021, *JASA*)
Elastic-net	$L_1 + L_2$ regression, sparse donor weights	Doudchenko & Imbens (2016, NBER WP)
XGBoost	Gradient-boosted trees, flexible nonlinear	Chen & Guestrin (2016, *KDD*)
Bayesian Ridge	Bayesian linear with normal prior, donor PIPs	Tipping (2001); Brodersen et al. (2015)

Range covered. Convex (no extrap) $\rightarrow$ augmented (ridge fix) $\rightarrow$ regression (sparse) $\rightarrow$ nonlinear $\rightarrow$ Bayesian (uncertainty).

Model setup

	Pre-event window (walk-forward CV)		Treatment (post)	
Russia 2022	2020-07-01 → 2022-02-23	421 obs	2022-02-24 → 2022-09-30	151 obs
Hormuz 2026	2024-06-03 → 2026-01-30	423 obs	2026-02-01 → 2026-05-01	∼59 obs

Hyperparameter tuning — multi-fold walk-forward CV.

- 5 expanding-window folds inside the pre-event period; 20-bday horizon per fold.
- Pick hparams by pooled val RMSE (Hyndman & Athanasopoulos 2018 §5.10).
- Final fit refits on full pre-event window, projects through treatment.

Limitation. No held-out test set beyond train/val; generalisation to unseen data cannot be measured directly. Formal inference → in-space placebo.

Validation

1. **Model fit.** Is the synthetic a credible pre-event approximator?
   - Walk-forward CV (Hyndman & Athanasopoulos 2018; Bergmeir & Benítez 2012).
2. **Donor identification.** Are the donors statistically clean of the event (SUTVA)?
   - Qualitative audit on causal channels (Abadie 2021 §3.1).
   - Bootstrap structural break at T_0 (Andrews 1993; Hansen 2000).
3. **Inference.** Is the post-event gap larger than under the no-treatment null?
   - In-space placebo, 21- and 33-donor pools (Abadie 2010 §3.3).
   - Cross-event weight transfer — Hormuz only, substitutes for the low-power placebo at $\sim$60 post-event obs.

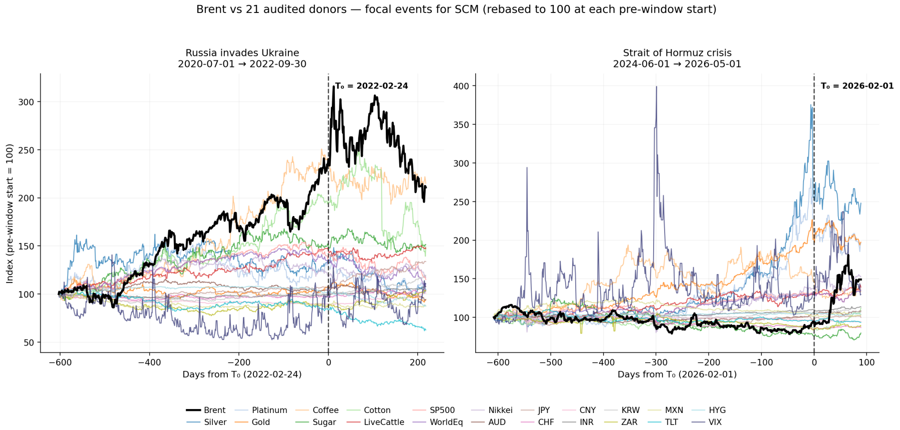
Donor selection

21 audited donors. Inclusion requires *both*: **pre-treatment** co-movement with Brent through a common global factor, *and* **treatment** cleanliness (no Russia/Ukraine or Persian Gulf exposure — SUTVA).

Category	Chosen donors	Factor channel (pre) & cleanliness (treatment)
Metals (3)	Silver, Platinum, Gold	Real rates / safe-haven / auto industry; no Russia or Gulf supply routing.
Agriculturals (4)	Coffee, Sugar, Cotton, LiveCattle	Global trade activity (Brazil / India / US producers); outside the Russia-Ukraine grain belt.
Equities (3)	SP500, WorldEq, Nikkei	Global growth / equity risk premium; no direct Russia or Gulf-energy exposure.
FX (8)	AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN	Safe-haven, EM growth, China demand; no petrocurrency or EU-energy-crisis loading.
Rates / credit (2)	TLT, HYG	US duration / credit-risk premia; not directly oil-linked.
Volatility (1)	VIX	S&P 500 implied vol — risk-on/off via equities.

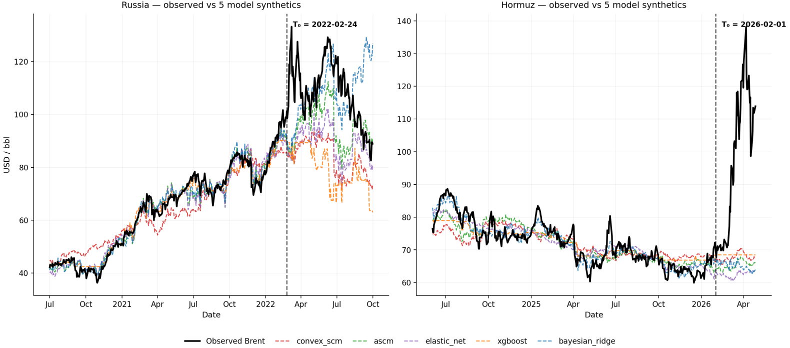
Categorical exclusions and per-event flag table: docs/donor_catalog.md.

Donor co-movement with Brent — audited pool



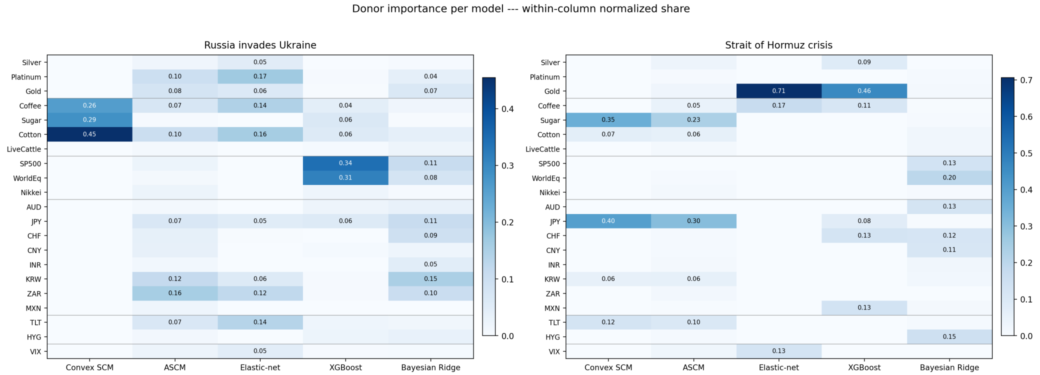
Brent (black) vs 21 audited donors, rebased to 100 at pre-window start; dashed line marks T_0. Same windows as the SCM.

Validation results — model fit



Model	Russia train	Russia val	Hormuz train	Hormuz val
Convex SCM	0.111	0.086	0.059	0.129
ASCM	0.046	0.087	0.035	0.087
Elastic-net	0.055	0.094	0.043	0.049
XGBoost	0.039	0.135	0.045	0.089
Bayesian Ridge	0.034	0.171	0.033	0.093

Donor importance



Per-model raw metric: convex weight w_j (SCM, ASCM) / $|\hat{\beta}_j|$ (Elastic-net, Bayesian Ridge) / feature importance (XGBoost). Cells = $|w_j|$ normalised within each column to sum to 1.

Validation results — in-space placebo

Abadie's canonical test (Abadie 2010 §3.3): rank Brent's post/pre RMSPE ratio against the placebo distribution.

Two donor pools for resolution.

- Shared 21-donor preferred: p-floor $= 1/22 \approx 0.045$.
- Full 33-donor pool: p-floor $= 1/34 \approx 0.029$.

Headline.

	Russia 2022	**Hormuz 2026**
21-donor	[p]	[p]
33-donor	[p]	[p]

Everything else (parallel-fit, regime-stability, in-time placebo, LOO, cross-event transfer): appendix.

External validation — EIA STEO pre-invasion forecasts (Russia only)

Cross-method check. SCM counterfactual vs EIA's last pre-invasion Brent forecast (STEO Feb-22, issued 2022-02-08). No shared information path (SCM uses cross-assetco-movement; STEO uses fundamental supply-demand modelling)

Model	Synth $/bbl	STEO Feb-22 $/bbl	Δ
Convex SCM	83.55	84.94	−$1.39
ASCM	95.52	84.94	+$10.58
Elastic-net	86.61	84.94	+$1.67
XGBoost	79.37	84.94	−$5.57
Bayesian Ridge	107.91	84.94	+$22.97 (deg.)
Ensemble median	**86.61**	**84.94**	**+$1.67**

Implied ATT (Brent actual $108.54): SCM ensemble +25.3% vs STEO Feb-22 +27.8% ($\Delta \sim 2.5$ pp).

Headline result — ATT estimate

ATT (% log Brent) over the post-event window.

Model	**Russia 2022** (2022-02-24 → 2022-09-30)	**Hormuz 2026** (2026-02-01 → latest)
Convex SCM	[%]	[%]
ASCM	[%]	[%]
Elastic-net	[%]	[%]
XGBoost	[%]	[%]
Bayesian Ridge	[%]	[%]
Ensemble median	**[%]**	**[%]**
IQR	[band]	[band]

Reading.

- Russia: calibration case; compare to documented \$95 → \$130+ move.
- Hormuz: main thesis result; place against scenario estimates (Dallas Fed, OIES, IEA).

Part 2 — ideas for the second half

[to be added by author]

Possible directions: pass-through to retail fuel / headline CPI; SCM vs scenario-model comparison; severity-dependent extrapolation; alternative donor pools.

Thank you

Questions and discussion

github.com/ElvisCasco/Brent_Analysis